Chapter 15 · Semester 2

DIRECT CURRENT CIRCUITS

Notes by Cikgu Suhaili

20 worked examples 3 simulations Exercises with answers

15.1 Electromotive force, potential difference and internal resistance

Electromotive force (e.m.f.), ε

of a source: the **energy converted from other forms (e.g. chemical) into electrical energy per unit charge** that passes through the source. It equals the p.d. across the terminals when no current flows (open circuit).

Potential difference, V

between two points: the **electrical energy converted into other forms per unit charge** passing from one point to the other.

$$\varepsilon = \frac{W}{Q} \qquad V = \frac{W}{Q}$$

both measured in volts (J C^{-1}); the difference is the direction of the energy change

	e.m.f.	potential difference
Energy change	other forms → electrical energy (in the source)	electrical energy → other forms (in a component)
Where	across a source (cell, generator)	between any two points in a circuit
Depends on the current?	no: a property of the source	yes (e.g. $V = IR$)

Internal resistance

Every real source has some resistance of its own, its **internal resistance** r (in a chemical cell: the electrolyte and the electrodes oppose the flow of charge). Part of the e.m.f. is used to drive the current through r, so the p.d. at the terminals is less than ε.

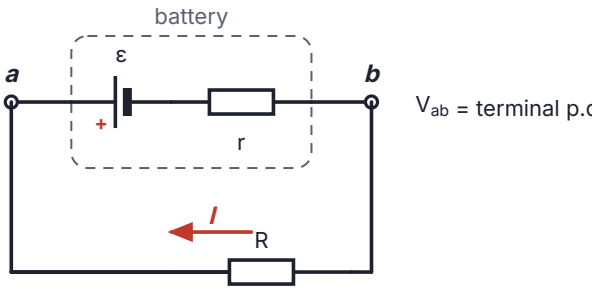


Figure 15.1: a cell of e.m.f. ε and internal resistance r connected to an external resistor R.

$$\varepsilon = I(R + r) \qquad V = \varepsilon - Ir$$

$V = IR$ = terminal p.d.; Ir = the "lost volts" across the internal resistance

- The larger the current drawn, the larger Ir and the lower the terminal p.d. (headlights dim when a car engine is started).
- $V = \varepsilon$ only when $I = 0$ (open circuit) or when $r = 0$ (an ideal source).
- When a cell is being **charged** (current forced into its + terminal), $V = \varepsilon + Ir$, which is larger than ε.

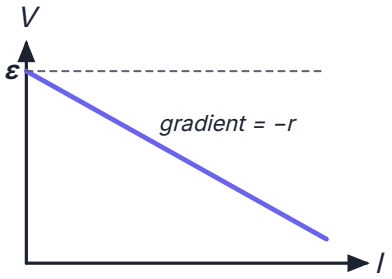
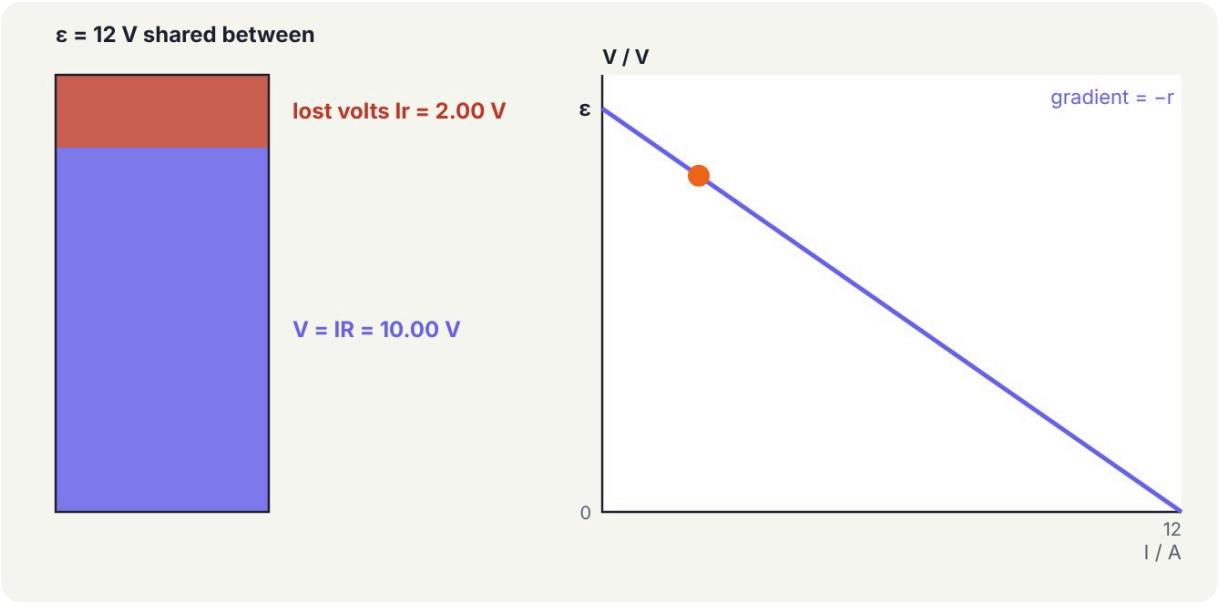


Figure 15.2: terminal p.d. against current: the intercept is ε and the gradient is −r.

Simulation

E.m.f. and internal resistance

Change the load resistance R and the internal resistance r. Watch the terminal p.d. fall as the current rises, and where the point lies on the V–I line.



current I = ε/(R + r)	terminal p.d. V	lost volts Ir	power to R	power wasted in r
2.00 A	**10.00 V**	**2.00 V**	**20.0 W**	**4.0 W**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 15 to use the simulation.

Example 15.1.1

A battery delivering 0.50 A has a terminal p.d. of 11.95 V and an internal resistance of 0.10 Ω. What is its e.m.f.?

Solution

$\varepsilon = V + Ir = 11.95 + (0.50)(0.10) =$ **12.0 V**

Example 15.1.2

A battery of e.m.f. 9.0 V is connected to a resistor. The current is 0.10 A and a voltmeter across the terminals reads 8.9 V. Find the internal resistance.

Solution

$$r = \frac{\varepsilon - V}{I} = \frac{9.0 - 8.9}{0.10} = \mathbf{1.0\ \Omega}$$

Example 15.1.3

When a cell gives a current of 0.20 A its terminal p.d. is 1.40 V; when it gives 0.60 A the terminal p.d. is 1.20 V. Find the e.m.f. and the internal resistance.

Solution

$1.40 = \varepsilon - 0.20r$ and $1.20 = \varepsilon - 0.60r$

Subtract: $0.20 = 0.40r \Rightarrow r = \mathbf{0.50\ \Omega}$; $\varepsilon = 1.40 + 0.10 = \mathbf{1.50\ V}$

Electrical energy and power

$$W = QV = VIt \qquad P = IV = I^2R = \frac{V^2}{R}$$

$P = IV$ for any device; I^2R and V^2/R only for a resistor (all the energy becomes heat).

Power supplied by the source = $I\varepsilon$.

Example 15.1.4

(a) Find the resistance of a 40 W, 12 V car headlamp. (b) A heater element of resistance 12 Ω carries 13 A. Find its power. (c) An iron with a 15 Ω element works at 120 V. How much heat does it produce in 1.0 h?

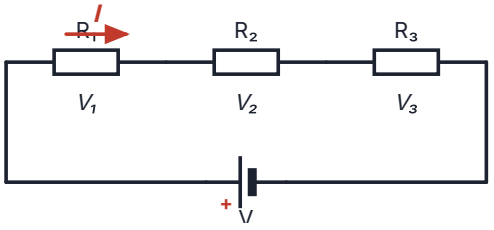
Solution

(a) $R = V^2/P = 12^2/40 =$ **3.6 Ω**

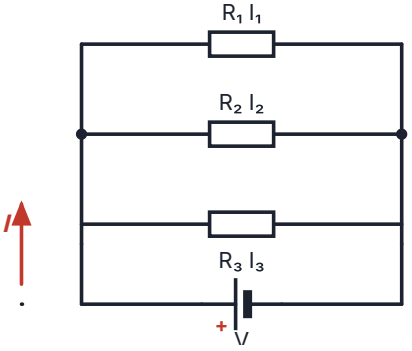
(b) $P = I^2R = (13)^2(12) =$ **2.0×10^3 W**

(c) $W = \dfrac{V^2}{R}t = \dfrac{120^2}{15}(3600) =$ **3.5×10^6 J**

Resistors in series and in parallel



Series: same current



Parallel: same p.d.

$$R_{eq} = R_1 + R_2 + R_3$$

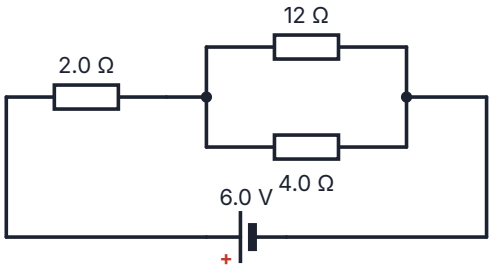
series: $V = V_1 + V_2 + V_3$ and $V = IR_{eq}$, so $IR_{eq} = IR_1 + IR_2 + IR_3$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

parallel: $I = I_1 + I_2 + I_3$ and each $I = V/R$, then divide by V

Example 15.1.5

Find (a) the total resistance, (b) the current from the 6.0 V supply, (c) the p.d. across the 4.0 Ω resistor.



Solution

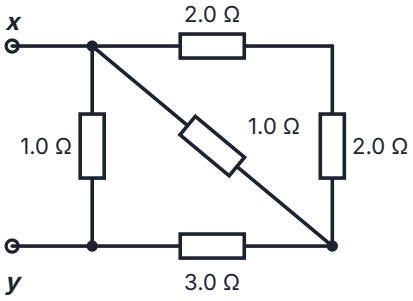
(a) 12 Ω ∥ 4.0 Ω: $\frac{(12)(4.0)}{16}$ = 3.0 Ω; R_{total} = 2.0 + 3.0 = **5.0 Ω**

(b) I = 6.0/5.0 = **1.2 A**

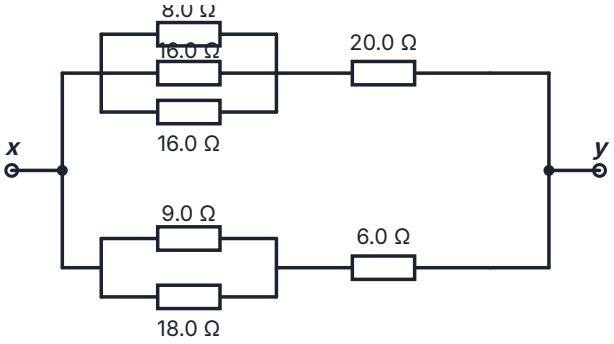
(c) $V_{2\Omega}$ = (1.2)(2.0) = 2.4 V, so $V_{4\Omega}$ = 6.0 − 2.4 = **3.6 V**

Example 15.1.6

Find the equivalent resistance between x and y in each network.



(a)



(b)

Solution

(a) 2.0 Ω + 2.0 Ω = 4.0 Ω (top and right side, in series) in parallel with the 1.0 Ω diagonal: 0.80 Ω; add the 3.0 Ω: 3.80 Ω; in parallel with the 1.0 Ω between x and y: $\frac{(3.80)(1.0)}{4.80}$ = **0.79 Ω**

(b) upper branch: 8.0 ‖ 16.0 ‖ 16.0 = 4.0 Ω, plus 20.0 Ω = 24.0 Ω. Lower branch: 9.0 ‖ 18.0 = 6.0 Ω, plus 6.0 Ω = 12.0 Ω. Together: $\frac{(24)(12)}{36}$ = **8.0 Ω**

15.2 Kirchhoff's laws

Kirchhoff's first law (junction rule)

The **sum of the currents entering a junction equals the sum of the currents leaving it** ($\Sigma I = 0$). It follows from the conservation of charge.

Kirchhoff's second law (loop rule)

In any closed loop, the **sum of the e.m.f.s equals the sum of the products IR** ($\Sigma \varepsilon = \Sigma IR$). It follows from the conservation of energy.



Figure 15.3: sign convention when going round a loop.

Mark a current (any direction you like) in each branch. Use the first law so the junctions balance.

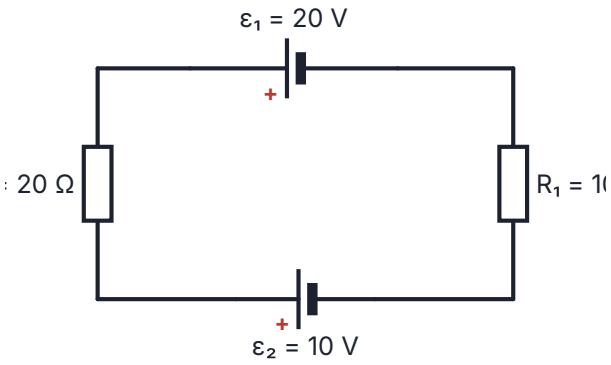
Choose a loop and a direction to go round it.

Apply $\Sigma\varepsilon = \Sigma IR$: an e.m.f. counts + if you pass through it from − to +; an IR counts + if you go the same way as the current you marked.

Solve the equations. A negative answer means that current really flows the other way; its size is still correct.

Example 15.2.1

Find the current in the circuit.



Solution

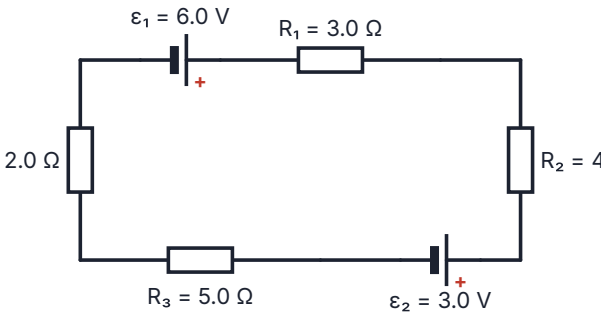
One loop, one current. The two cells oppose each other; ε_1 is larger, so take I anticlockwise (driven by ε_1).

$\Sigma\varepsilon = \Sigma IR$: $20 - 10 = I(20 + 10)$

$I = 10/30 =$ **0.33 A** in each resistor

Example 15.2.2

Find the current in each resistor.



Solution

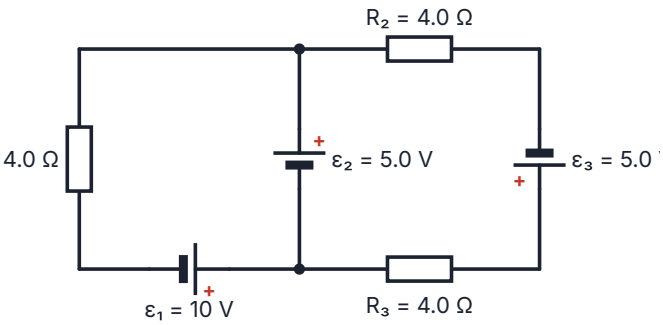
One loop. Take I clockwise. ε_1 is passed from − to + (+6.0 V); ε_2 is passed from + to − (−3.0 V).

$6.0 - 3.0 = I(3.0 + 4.0 + 5.0 + 2.0) = 14I$

I = **0.21 A**, clockwise, the same in every resistor

Example 15.2.3

Find the current in each branch.



Solution

Let I_1 flow through R_1 and ε_1, I_3 through R_2, ε_3 and R_3, and I_2 up through ε_2. Junction: $I_2 = I_1 + I_3$.

Left loop (ε_1, ε_2, R_1): $10 + 5.0 = 4.0 I_1 \Rightarrow I_1 =$ **3.75 A**

Right loop (ε_2, ε_3, R_2, R_3): $5.0 + 5.0 = (4.0 + 4.0) I_3 \Rightarrow I_3 =$ **1.25 A**

$I_2 = 3.75 + 1.25 =$ **5.0 A**

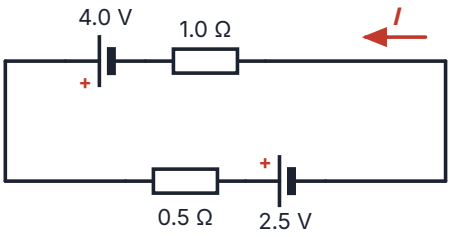
Example 15.2.4

A cell of e.m.f. 4.0 V and internal resistance 1.0 Ω is connected in series with a cell of e.m.f. 2.5 V and internal resistance 0.5 Ω in a closed loop, so that the current is as small as possible. Draw the circuit and find the current.

Solution

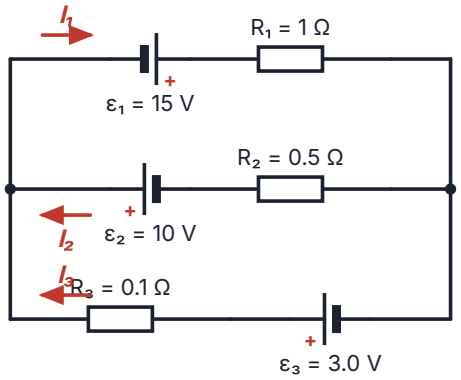
The current is smallest when the cells **oppose** each other: connect + to + (and − to −).

$$I = \frac{4.0 - 2.5}{1.0 + 0.5} = \mathbf{1.0\ A}$$



Example 15.2.5

Find I_1, I_2 and I_3. Ignore the internal resistances.



Solution

Junction (right-hand node): $I_1 = I_2 + I_3$ ... (1)

Top loop (branches 1 and 2): $15 + 10 = (1)I_1 + (0.5)I_2 \Rightarrow 25 = I_1 + 0.5I_2$ ... (2)

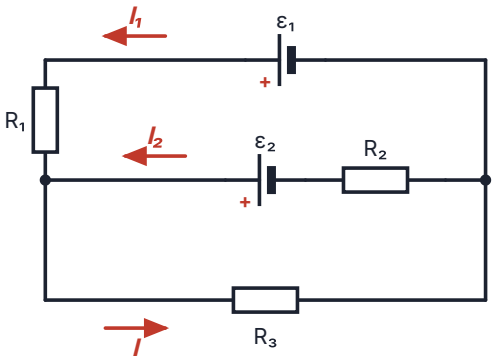
Bottom loop (branches 2 and 3): $3.0 - 10 = 0.1I_3 - 0.5I_2 \Rightarrow -7 = 0.1I_3 - 0.5I_2$ ... (3)

From (3): $I_3 = 5I_2 - 70$. Put into (1) and (2): $25 = 1.5I_2 + 5I_2 - 70 \Rightarrow I_2 =$ **14.6 A**

$I_3 = 5(14.62) - 70 =$ **3.1 A**; $I_1 = 14.62 + 3.08 =$ **17.7 A**

Example 15.2.6

ε_1 = 8 V, R_1 = 1 Ω, R_2 = 2 Ω, R_3 = 3 Ω and I = 3 A. Ignore the internal resistances. Find (a) I_1 and I_2, (b) ε_2.



Solution

Loop through ε_1, R_1 and R_3: with the directions marked, $0 = 1(I_1) - 8 + 3(3) \Rightarrow I_1 = -1$ A.

(a) I_1 **= 1 A**, flowing in the **opposite** direction to the arrow. Junction: $I = I_1 + I_2 \Rightarrow 3 = -1 + I_2 \Rightarrow I_2$ **= 4 A**

(b) Loop through ε_2, R_2 and R_3: $\varepsilon_2 = I_2R_2 + IR_3 = (4)(2) + (3)(3) =$ **17 V**

15.3 Potential divider

Two resistors in series across a supply share the supply voltage in the ratio of their resistances. Taking the output across one of them gives a smaller, chosen voltage.

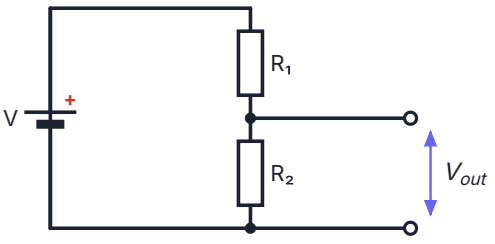


Figure 15.4: a potential divider.

$$V_{\text{out}} = \frac{R_2}{R_1 + R_2} V$$

the same current $I = V/(R_1 + R_2)$ flows through both resistors (no load connected)

A source of variable voltage

If the two resistors are replaced by one uniform resistance wire (or a rheostat) with a sliding contact, the output can be varied smoothly from 0 to the full supply voltage. For a uniform wire, $R \propto l$, so $V_{\text{out}} = \frac{l}{L} V$.

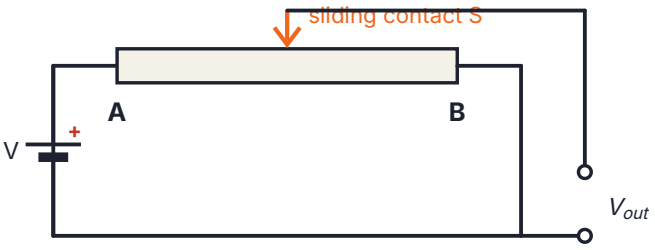


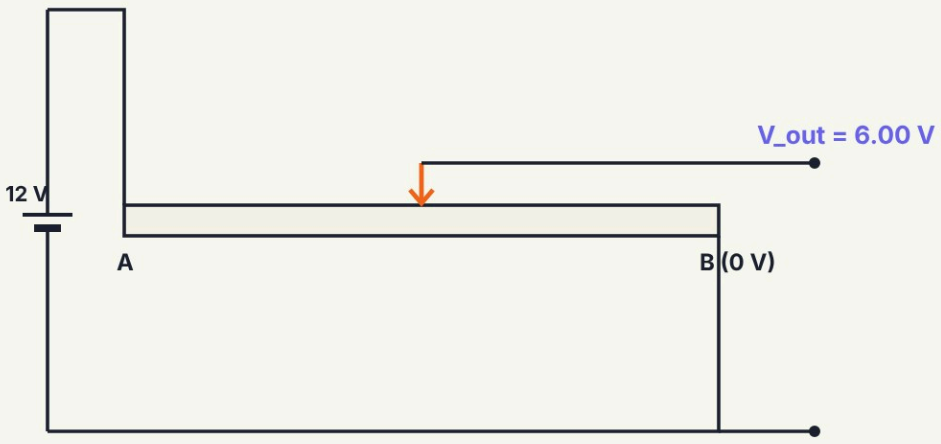
Figure 15.5: a rheostat used as a potential divider (potentiometer connection).

If a load is connected across the output, it is in parallel with R_2. The combined resistance is smaller, so V_{out} falls. The output only stays near the calculated value if the load resistance is much larger than R_2.

Simulation

Potential divider

Move the sliding contact to change the output voltage. Then connect a load and make it smaller: watch the output fall below the value without a load.



output (no load)	output now	lower part of the rheostat	load
6.00 V	6.00 V	0.50 kΩ (total 1 kΩ)	not connected

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 15 to use the simulation.

Example 15.3.1

Resistors of 3.0 Ω and 6.0 Ω are connected in series to a 12.0 V battery of negligible internal resistance. Find the p.d. across each resistor.

Solution

$V_3 = \frac{3.0}{9.0}(12.0) = \mathbf{4.0\ V}$; $V_6 = \frac{6.0}{9.0}(12.0) = \mathbf{8.0\ V}$

Example 15.3.2

A 12 V supply is connected across two 10 kΩ resistors in series. Find the output across the lower resistor (a) with no load, (b) with a 10 kΩ load connected across it.

Solution

(a) $V_{out} = ½(12) = \mathbf{6.0\ V}$

(b) 10 kΩ ∥ 10 kΩ = 5.0 kΩ: $V_{out} = \frac{5.0}{10 + 5.0}(12) = \mathbf{4.0\ V}$

Shunts and multipliers

A galvanometer has a resistance R_g and gives full-scale deflection for a small current I_g. It can be converted into an ammeter or a voltmeter.



Figure 15.6: a shunt in parallel makes an ammeter; a multiplier in series makes a voltmeter.

Ammeter: shunt

- A small resistance R_s is connected in **parallel** with the galvanometer so that most of the current bypasses it.
- Same p.d.: $I_g R_g = (I - I_g) R_s$

$$R_s = \frac{I_g R_g}{I - I_g}$$

an ammeter should have a very small resistance

Voltmeter: multiplier

- A large resistance R_m is connected in **series** with the galvanometer.
- At full scale: $V = I_g(R_g + R_m)$

$$R_m = \frac{V}{I_g} - R_g$$

a voltmeter should have a very large resistance

Example 15.3.3

A galvanometer of resistance 50 Ω gives full-scale deflection for 1.0 mA. How can it be converted into (a) an ammeter reading up to 1.0 A, (b) a voltmeter reading up to 10 V?

Solution

(a) $R_s = \frac{(1.0 \times 10^{-3})(50)}{1.0 - 0.0010}$ = **0.050 Ω**, connected in parallel

(b) $R_m = \frac{10}{1.0 \times 10^{-3}} - 50$ = **9950 Ω**, connected in series

15.4 Potentiometer and Wheatstone bridge

Potentiometer

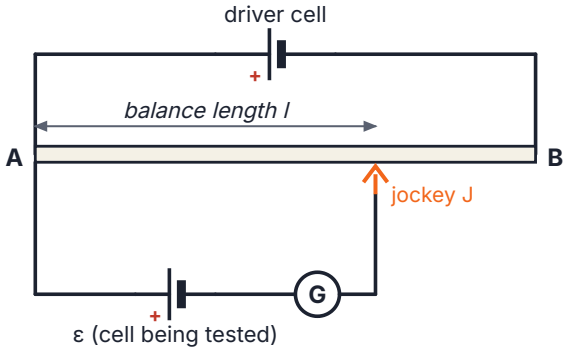


Figure 15.7: a potentiometer measuring the e.m.f. of a cell.

- A **driver cell** sends a steady current through a uniform resistance wire AB (often 100 cm long). The p.d. along the wire is proportional to length: $V_{AJ} \propto l$.

- The cell being tested is connected with its + terminal to A (the same side as the + of the driver cell), through a galvanometer to a sliding **jockey** J.
- The jockey is moved until the galvanometer shows **no deflection** (balance). Then no current flows through the test cell, so the p.d. across length l equals its **e.m.f.** exactly.

$$\varepsilon = \frac{l}{L} V_{AB} \qquad \frac{\varepsilon_1}{\varepsilon_2} = \frac{l_1}{l_2} \qquad r = R\frac{l_0 - l}{l}$$

measuring an e.m.f.; comparing two e.m.f.s; internal resistance (l_0 = balance length on open circuit, l = balance length with a resistor R across the cell)

- **Uses:** measuring an e.m.f. (no current is drawn, unlike a voltmeter), comparing e.m.f.s, measuring internal resistance, comparing resistances.
- **No balance point** (deflection one way all along the wire)? The e.m.f. being measured is larger than the p.d. across the whole wire, or the cells are connected + to −, or a connection is loose.

Wheatstone bridge

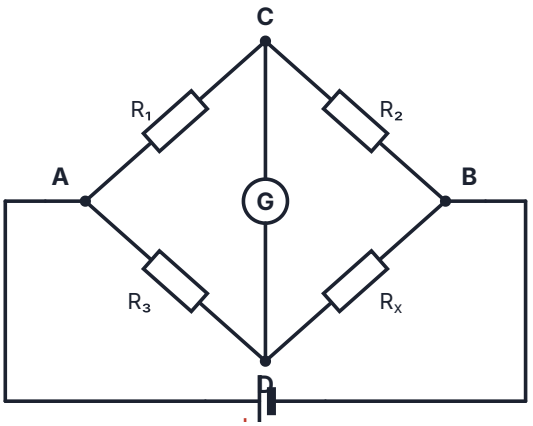


Figure 15.8: a Wheatstone bridge.

Four resistors form a bridge with a galvanometer between C and D. One of them, R_x, is unknown.

A known resistor is adjusted until the galvanometer reads zero: the bridge is **balanced**, so C and D are at the same potential.

No current in CD means R_1 and R_2 carry the same current I_1, and R_3 and R_x carry I_2.

$V_{AC} = V_{AD}$: $I_1R_1 = I_2R_3$; $V_{CB} = V_{DB}$: $I_1R_2 = I_2R_x$. Divide the two equations.

$$\frac{R_1}{R_2} = \frac{R_3}{R_x} \Rightarrow R_x = \frac{R_2 R_3}{R_1}$$

balance does not depend on the e.m.f. of the cell or on the galvanometer resistance

Metre bridge

A practical Wheatstone bridge: two arms are the two parts of a uniform wire 100 cm long. At balance, the ratio of resistances equals the ratio of lengths.

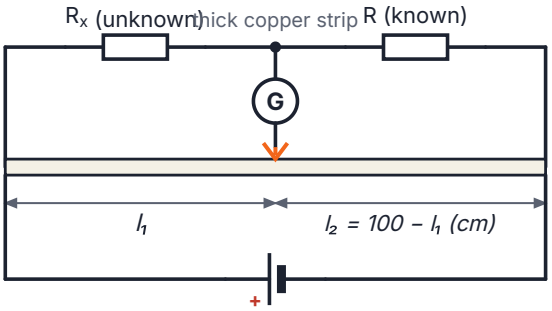


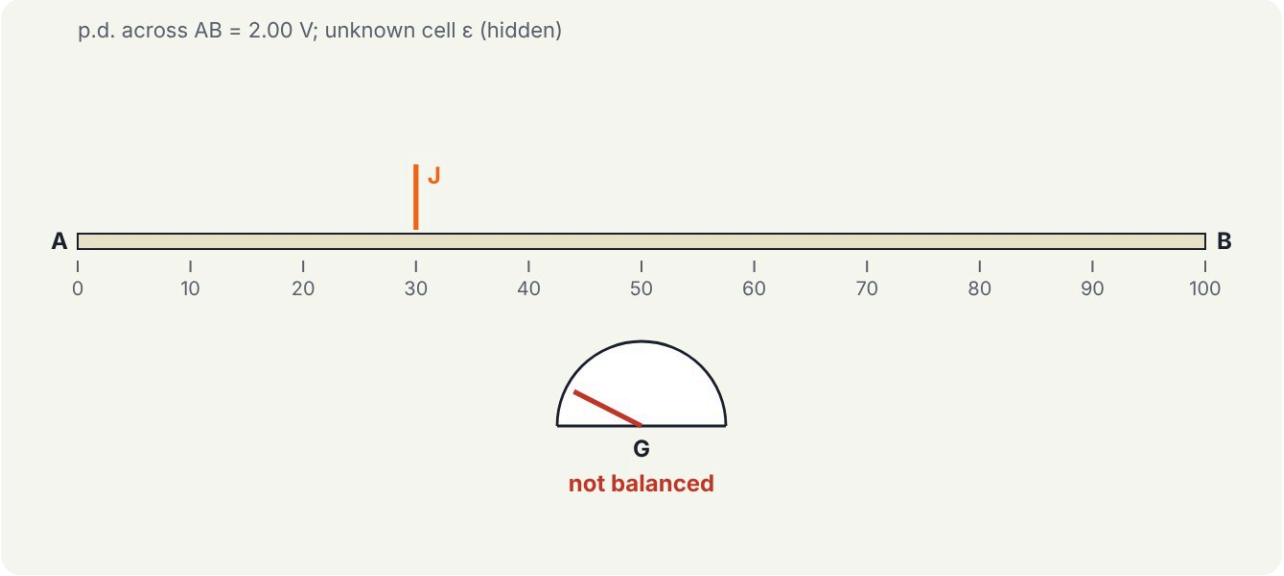
Figure 15.9: a metre bridge.

$$\frac{R_x}{R} = \frac{l_1}{l_2}$$

best accuracy when the balance point is near the middle of the wire

Simulation Potentiometer and metre bridge

Slide the jockey along the wire until the galvanometer reads zero. Read the balance length and use it to find the unknown e.m.f. or the unknown resistance.



balance length l
—

ε = (l/100) × 2.00 V
find the balance first

hint
move J towards B

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 15 to use the simulation.

Example 15.4.1

A potentiometer wire 100.0 cm long has a p.d. of 2.00 V across it. A cell balances at 75.0 cm. Find its e.m.f.

Solution

$$\varepsilon = \frac{75.0}{100.0}(2.00) = \mathbf{1.50\ V}$$

Example 15.4.2

A standard cell of e.m.f. 1.018 V balances at 60.0 cm on a potentiometer. Another cell balances at 72.0 cm. Find its e.m.f.

Solution

$$\varepsilon_2 = \varepsilon_1 \frac{l_2}{l_1} = 1.018 \times \frac{72.0}{60.0} = \mathbf{1.22\ V}$$

Example 15.4.3

A cell balances at 80.0 cm on open circuit. When a 10 Ω resistor is connected across it, the balance length becomes 64.0 cm. Find the internal resistance of the cell.

Solution

$$r = R\frac{l_0 - l}{l} = 10 \times \frac{80.0 - 64.0}{64.0} = \mathbf{2.5\ \Omega}$$

(Open circuit: $\varepsilon \propto l_0$. With R: $V = IR \propto l$. Since $\varepsilon/V = (R + r)/R = l_0/l$.)

Example 15.4.4

A platinum wire of diameter 0.920 mm is the unknown resistance in a Wheatstone bridge. R_1 = 38.0 Ω and R_2 = 46.0 Ω, and balance is reached when R_3 = 3.48 Ω. Find the length of the wire. ($\rho = 10.6 \times 10^{-8}$ Ω m)

Solution

$$R_x = \frac{R_2 R_3}{R_1} = \frac{(46.0)(3.48)}{38.0} = 4.21\ \Omega$$

$$l = \frac{R_x A}{\rho} = \frac{(4.21)\pi(0.920 \times 10^{-3})^2/4}{10.6 \times 10^{-8}} = \mathbf{26.4\ m}$$

Example 15.4.5

In a metre bridge the unknown resistor is in the left gap and a 10.0 Ω resistor in the right gap. Balance is at 40.0 cm from the left end. Find the unknown resistance.

Solution

$$R_x = R\frac{l_1}{l_2} = 10.0 \times \frac{40.0}{60.0} = \mathbf{6.67\ \Omega}$$

Exercise: d.c. circuits

1. A battery of e.m.f. 12 V and internal resistance 0.50 Ω is connected to a 5.5 Ω resistor. Find (a) the current, (b) the terminal p.d., (c) the power lost in the battery.

   Answer

   (a) 2.0 A (b) 11 V (c) I^2r = 2.0 W

2. Why does a potentiometer measure the e.m.f. of a cell more accurately than a voltmeter?

   Answer

   At balance no current flows through the cell, so there is no p.d. across its internal resistance. A voltmeter draws a small current, so it reads ε − Ir, slightly less than ε.

3. A galvanometer of resistance 20 Ω gives full-scale deflection with 5.0 mA. Find the resistance needed to make (a) a 0–5.0 A ammeter, (b) a 0–15 V voltmeter, and say how each is connected.

   Answer

   (a) 0.020 Ω in parallel (b) 2980 Ω in series